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Co-funded by the European Union
OPEN ASIA
UNIFYING SCIENCE, EMPOWERING INNOVATION
Project Number : 101128493

Localising Open Science: UNIMAS's Push for Digital Innovation and Excellence

Professor Dr. Wan Hashim Wan Ibrahim
Universiti Malaysia Sarawak



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A PARADIGM SHIFT

From institutional silos to Open Science

BEFORE

Research data buried in

- Private hard drives
- Proprietary databases
- Forgotten lab notebooks

→

TODAY

Accessible to all levels of society

- Research
- Publications
- Primary data
- Digital workflows

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A transformative national movement

01 National Science, Technology & Innovation Policy 2021-2030

02 Driven by
Ministry of Science, Technology and Innovation

03 Established
Malaysia Open Science Platform

04 Spearheaded by
Academy of Sciences Malaysia



KEMENTERIAN SAINS, TEKNOLOGI DAN INOVASI
MINISTRY OF SCIENCE, TECHNOLOGY AND INNOVATION

Open SCIENCE
Malaysia Open Science Platform (MOSP)



AKADEMI SAINS MALAYSIA

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A Clear Mandate



Transform public research data into a strategic national asset by building trusted, interconnected platforms that align with international best practices.

Maximise resource efficiency by enabling researchers across the nation to share high-value assets and form transdisciplinary research networks.

Prevent unnecessary research duplication so that public funding generates higher outputs and solves pressing socio-economic challenges.

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UNIMAS: The institutional engine for PCDS 2030



Core enablers realised through Open Science

- Digital Transformation
- Innovation
- Human Capital Development

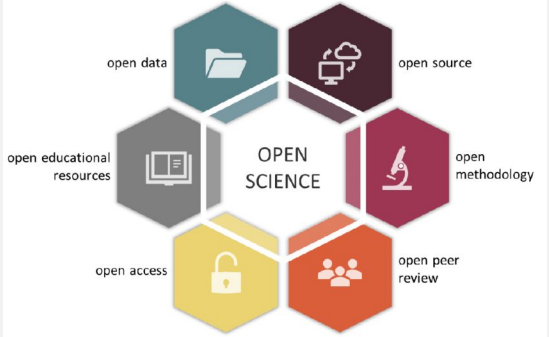
Not merely participating, **championing Open Science.**

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Balancing global and local

GLOBAL MANDATE

Open transparency





Unique biodiversity



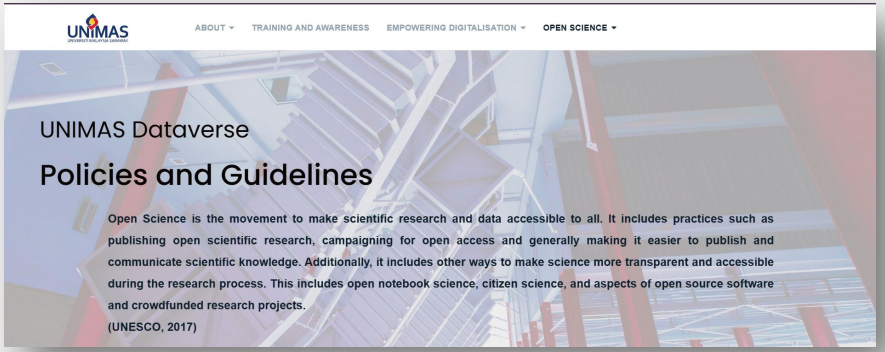
Indigenous heritage



Community rights

Institutional Governance

Open Science Research Data Management Policy




Universiti Malaysia Sarawak
(UNIMAS)
Open Science
Research Data Management
Policy

Scope of Application

All campus operations and all UNIMAS researchers — academic staff and postgraduate students.

The entire data lifecycle

Before | **During** | **After**

Data Ownership

All research data generated by UNIMAS personnel is wholly owned by the university — and remains with UNIMAS even if an academic departs.

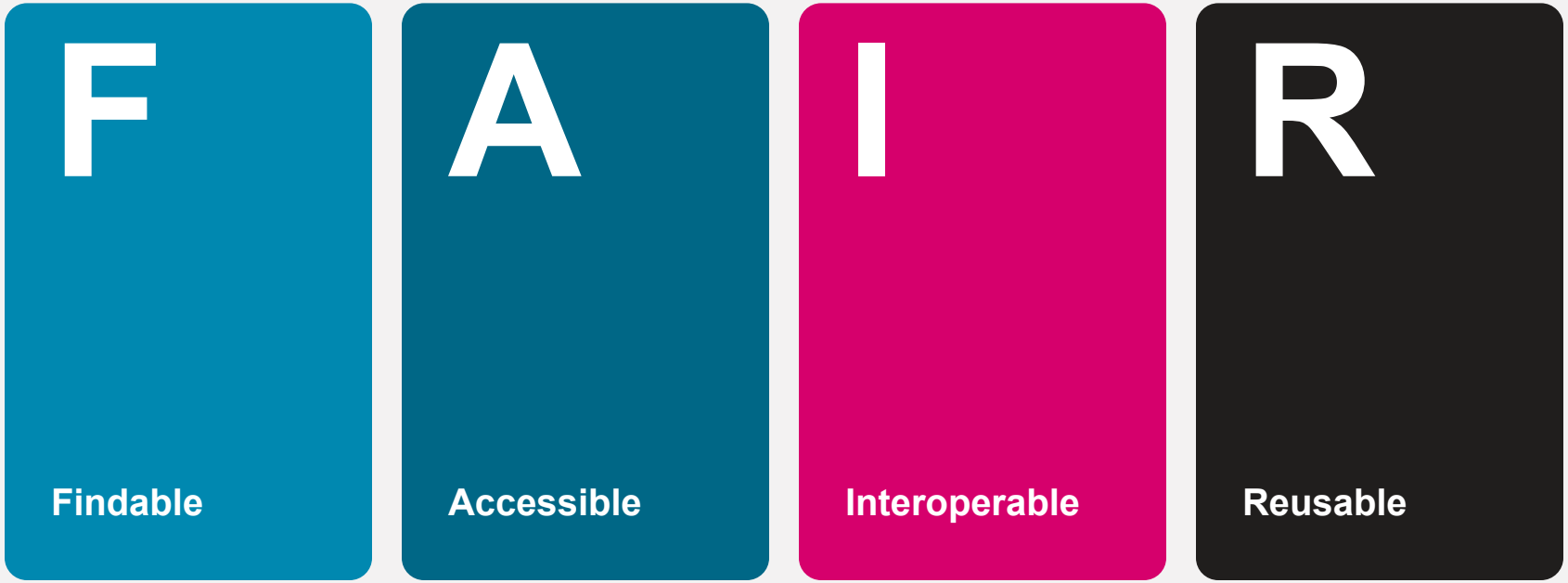
Exception: formal research funding contracts that explicitly stipulate the external funder retains data ownership.

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AT THE CENTRE OF THE POLICY

The FAIR Data Principles



FAIR IN PRACTICE

1 Open Science Data Management Plan

Submitted online by Principal Investigators **before** a single piece of data is gathered. It maps out:

- Collection purpose
- File formats
- Metadata descriptions
- Standard vocabularies
- Storage security
- Resource allocation
- Licensing for reuse

2 Mandatory Repository Deposition

Final data validating published findings goes into the **UNIMAS Open Science Data Repository (Dataverse)** no later than the article's publication date.

UNIMAS Dataverse

The Dataverse® Project

Gold Open Access
Immediate public access

Green Open Access
Institutional self-archiving

Three key internal centres

UNIMAS UNIVERSITI MALAYSIA SARAWAK | TUN ABANG HAJI OPENG DIGITAL CENTRE

TAHODC

Tun Abang Haji Openg Digital Centre

Technical backbone

Secure servers, Dataverse maintenance, automated backups and system monitoring.

UNIMAS UNIVERSITI MALAYSIA SARAWAK | UNIMAS INNOVATION

RDICE

Centre for Research, Development, Innovation, Commercialisation & Economy

Administrative engine

Policy implementation, software subscriptions, compliance monitoring and grant applications.



PeTARY

Perpustakaan Tun Abdul Rahman Ya'kub

Data stewardship

Metadata catalogue, hands-on FAIR training, and advice on open access, metadata standards and curation.

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THE GUIDING PHILOSOPHY

"As open as possible, as closed as necessary."

SOP Standard Operating Procedure for Handling Ethical and Sensitive Data in Open Science — balancing transparency with ethical and legal safeguards.

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Standard Operating Procedure (SOP) for Handling Ethical and Sensitive Data in Open Science

A 5-tier classification framework

1	**Personal Data**	Identifiable individuals; governed by the Personal Data Protection Act 2010 (Act 709).
2	**Confidential Data**	Trade secrets, active investigations, financial security and national safety matters.
3	**Restricted Data**	Shared only with verified, authorised user groups under controlled access.
4	**Proprietary Interest Data**	Commercialisation potential; not shared until the patent application is officially filed.
5	**National Security Data**	Cleared by authorised security bodies and fully restricted.

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LEGISLATIVE GUARDRAILS

Protecting our unique environment and heritage

2000

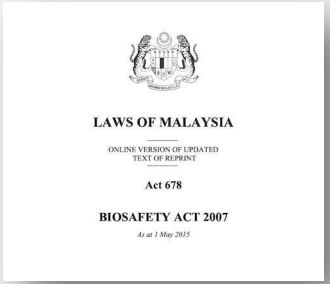
Sarawak Biodiversity Enactment

2003

Sarawak Biodiversity Centre Ordinance (Amendment)

2007

National Biosafety Act


LAWS OF MALAYSIA

ONLINE VERSION OF UPDATED TEXT OF REPRINT

Act 678

BIOSAFETY ACT 2007

As at 1 May 2015

2020

Malaysian Code of Responsible Conduct in Research (MCRCR), ASM


THE MALAYSIAN CODE OF RESPONSIBLE CONDUCT IN RESEARCH

2nd Edition

NATIONAL SCIENCE COUNCIL

Open research and responsible governance can coexist in full harmony.

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Our campus digital ecosystem

01 UNIMAS Digital Experience Hub

Malaysia's first public university digital experience hub, in strategic collaboration with Google for Education.















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Our campus digital ecosystem (cont'd)

02

Google Workspace for Education

Cloud collaboration and secure data exchange across faculties and international research teams.







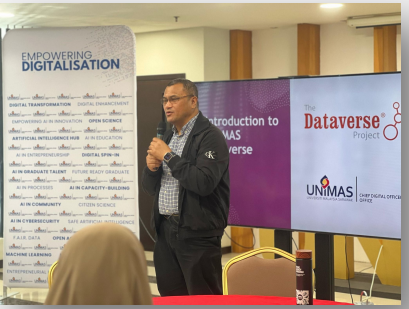


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Our campus digital ecosystem (cont'd)

03 Digital Transformation Seminar Series

Continuous training in digital tools, data stewardship and cloud analytics.















Our campus digital ecosystem (cont'd)

04 The Open Asia project

International networks for open science practice, cross-border collaboration and shared data standards.











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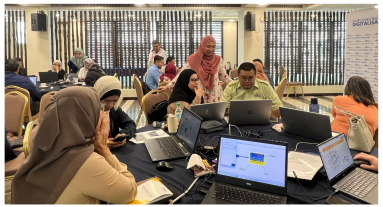
A fundamental cultural transformation

The Dataverse Project

Seminar Series workshops train our research community to

- Navigate the UNIMAS Dataverse
- Structure a compliant OSDMP
- Apply standard creative licences
- Publish in high-impact open access repositories





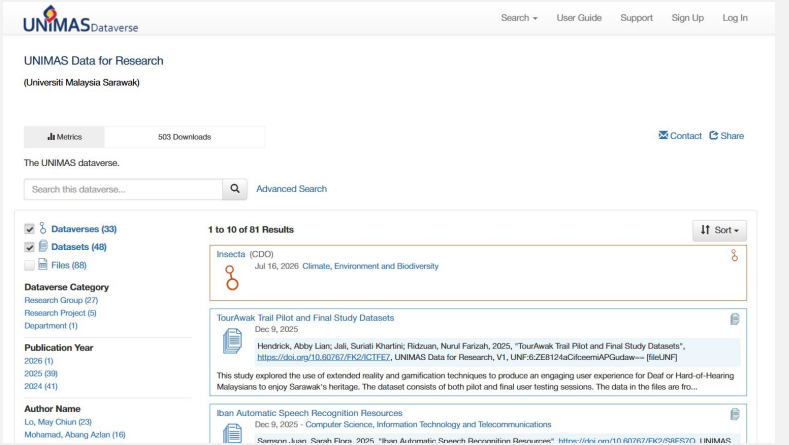


MOVING AWAY FROM

Data as private property, guarded in isolation

MOVING TOWARD

Well-curated data as a public good



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TALENT FOR PCDS 2030

A tech-savvy research workforce for Sarawak's data-driven economy

Science, Technology, Engineering, and Mathematics (STEM)



Artificial Intelligence (AI)



Advanced technologies



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What Open Science means for Sarawak

Digital economy

Connectivity

Research transparency

Institutional trust

Local sovereignty

Digital excellence

TO RESEARCHERS

Treat OSDMPs not as a bureaucratic exercise, but as the foundational design of your research quality.

TO INSTITUTIONAL LEADERS

Drive the cultural shift: encourage open access publishing, support postgraduates in data stewardship, and celebrate data sharing.


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Thank you.